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EST II — MATH LEVEL 2 • UNIT 1.01

Complex Numbers, Modulus & Polar Form

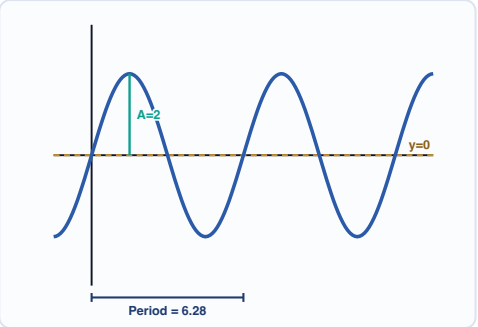
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Complex Numbers, Modulus & Polar Form Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?



A $2 + 3i$

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

2 MCQ

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

A 5

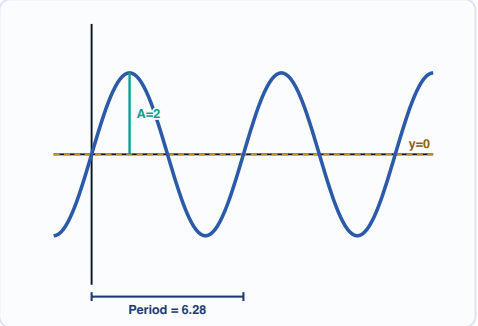
B 7

C 12

D 25

3 MCQ

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.



A 5

B 6

C 7

D 8

4 MCQ

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

A $\frac{17}{2}$

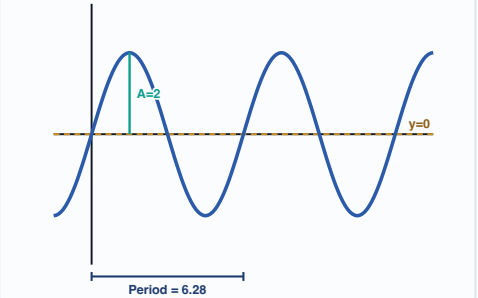
B $-\frac{17}{2}$

C $\frac{11}{2}$

D $-\frac{11}{2}$

5 MCQ

Let $z = 3 - 4i$. What is the real part of z?



A 3

B -4

C 5

D 7

6 MCQ

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

A 6

B 8

C 10

D 12

7 MCQ

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

A 5

B 7

C 12

D 25

8 MCQ

What is i^2?

A -1

B 1

C i

D $-i$

9 MCQ

Express in simplest algebraic form $(9+i)+(-7+i)$.

A 3+2i

B 2+2i

C 2+4i

D 16

10 MCQ

Express in simplest algebraic form $(2-9i)-(1+7i)$.

A 2-16i

B 1-14i

C 1-16i

D 3-2i

11 MCQ

Let the complex number $x = 24 + 14i$ have complex conjugate $\bar{x}$ on the Argand diagram. What is the squared modulus $f = x\bar{x} = |x|^2$?

A 796

B 758

C 772

D 634

12 MCQ

Express in simplest algebraic form i^{26}.

A 1

B i

C -1

D -i

13 MCQ

Rewrite in fully simplified form i^{21}.

A i

B -i

C 1

D -1

14 MCQ

Reduce to its simplest form i^7.

A 1

B i

C -1

D -i

15 MCQ

Express in simplest algebraic form i^{34}.

A -1
B 1
C i
D -i

16 MCQ

Rewrite in fully simplified form i^{3}.

A 1
B i
C -1
D -i

17 MCQ

Reduce to its simplest form i^{9}.

A i
B 1
C -1
D -i

18 MCQ

Express in simplest algebraic form i^{8}.

A i
B -1
C 1
D -i

19 MCQ

Rewrite in fully simplified form i^{30}.

A -1
B 1
C i
D -i

20 MCQ

Express in simplest algebraic form $\frac{-5-5i}{5+5i}$.

A -1
B -i
C 0
D 1

21 MCQ

Express in simplest algebraic form $\frac{1+7i}{2-i}$.

A -1+3i

B 3-i

C 3i

D 1+3i

22 MCQ

Express in simplest algebraic form $\frac{-2+6i}{2-6i}$.

A -i

B -1

C 0

D 1

23 MCQ

Rewrite in fully simplified form $\frac{-7-9i}{1+2i}$.

A 1-5i

B -5+i

C -4+i

D 5+i

24 MCQ

Simplify $(5+2i)+(3-4i)$.

A $8+2i$

B $8-2i$

C $2-2i$

D $8-6i$

25 MCQ

Simplify $(7-3i)-(2-i)$.

A $5-2i$

B $5-4i$

C $9-2i$

D $5+2i$